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ARTICLE

Steady and unsteady pressure scour under bridges at clear-water conditions

Serife Yurdagul Kumcu

Abstract: Bridges are vital components of the transportation network and if they are damaged or destroyed during a flood, they isolate communities and limit movement of supplies and emergency services. Evaluating their constructional stability and structural response after a flood event is critical for bridge safety. Bridge studies are usually designed with an assumption of an open channel flow condition, but the flow regime can switch to pressure flow if the downstream edge of a bridge deck is partially or totally submerged during a large flood. The main goal of this paper is to study the pressurized flow scour under a bridge deck and downstream deposition that results from eroded sediment material governed by both steady and unsteady clear-water flow conditions. Experimental conditions used in this study involve clear-water scour of a sand bed of given median sediment size d_{50} = 0.90 mm and sediment uniformity σ_g = 1.29, an approach flow characterized by a flow depth and velocity, a rectangular-shaped bridge deck, and a stepwise flood hydrograph defined by its time to peak and peak discharge. Different flow conditions were considered in confined flow under the bridge deck. Relationship between pressure-flow scour and flow conditions is presented and discussed under the obtained experimental data. Additionally, effects of single-peaked stepwise flow hydrographs (unsteady flow conditions) on bridge pier scour depth are investigated under clear-water pressure-flow conditions, whereas previous researches mainly focused on the equilibrium pressure scour under steady flow conditions.

Key words: pressured scour, clear-water flow, unsteady flow, temporal development, sediment transport.

Résumé : Les ponts constituent des éléments essentiels du réseau de transport et s'ils sont détruits ou endommagés lors d'une inondation, les collectivités s'en trouvent isolées et il y a des limitations quant au transport des approvisionnements et à la prestation des services d'urgence. Il est crucial pour assurer la sécurité des ponts d'évaluer leur stabilité de construction et leur réponse structurale à la suite d'un événement d'inondation. On conçoit habituellement les études des ponts selon l'hypothèse de la condition d'écoulement à surface libre, cependant le régime d'écoulement peut devenir un écoulement sous pression si le côté en aval du tablier du pont est partiellement ou complètement submergé au cours d'une importante inondation. L'objectif principal de cette étude est d'examiner l'affouillement dû à l'écoulement sous pression sous le tablier d'un pont et les dépôts en aval découlant de matériaux de sédiments érodés dépendant à la fois de conditions d'écoulement régulier et irrégulier d'eau claire. Les conditions expérimentales utilisées dans le cadre de cette étude se rapportent à l'affouillement d'eau claire d'un lit de sable de sédiments de taille médiane donnée de d_{50} = 0,90 mm et d'uniformité donnée de σ_g = 1,29, un écoulement d'approche ayant une profondeur et une vitesse d'écoulement distinctes, un tablier de pont rectangulaire et un hydrogramme de l'écoulement séquentiel déterminé par le temps nécessaire pour atteindre le débit de pointe et par le débit de pointe. On a considéré différentes conditions pour l'écoulement confiné sous le tablier du pont. On présente et analyse la relation entre l'affouillement de la pression d'écoulement et les conditions d'écoulement à la lumière des données expérimentales. De plus, on examine les effets d'hydrogrammes d'écoulement à pointe unique avec le temps (conditions d'écoulement irrégulier) sur la profondeur de l'affouillement des piles du pont sous des conditions de pression d'écoulement d'eau claire, tandis que les recherches antérieures portaient sur l'affouillement de pression d'équilibre sous des conditions d'écoulement régulier. [Traduit par la Rédaction]

Mots-clés : affouillement sous pression, écoulement d'eau claire, écoulement irrégulier, évolution dans le temps, transport de sédiment.

Introduction

Pressurized flow conditions under a bridge structure can be initiated if the approach flow depth rises above the bridge deck. Unlike open channel flows, pressure-flows create a severe scourability potential because scouring the channel bed is one of the only ways to dissipate the energy when passing a given discharge in pressurized flow (Guo et al. 2009). Previous research on pressurized flow scour is limited and includes studies by Abed (1991), Jones et al. (1993), Arneson (1997), Umbrell et al. (1998), Lyn (2008), Guo et al. (2009), Hahn and Lyn (2010), Zhai (2010), and Melville (2014). Based on the study of Arneson (1997), an equation named HEC-18 states

$$\frac{z_s}{h_a} = -5.08 + 1.27\left(\frac{h_a}{h_b}\right) + 4.44\left(\frac{h_a}{h_b}\right)^{-1} + 0.19\left(\frac{V_b}{V_c}\right) \quad (1)$$

where z_s is the depth of equilibrium or ultimate scour; h_a is the upstream flow depth; h_b is the initial (prior to scour) distance between the bridge low chord and the undisturbed bed at the bridge section; V_b is the initial (prior to scour) velocity through

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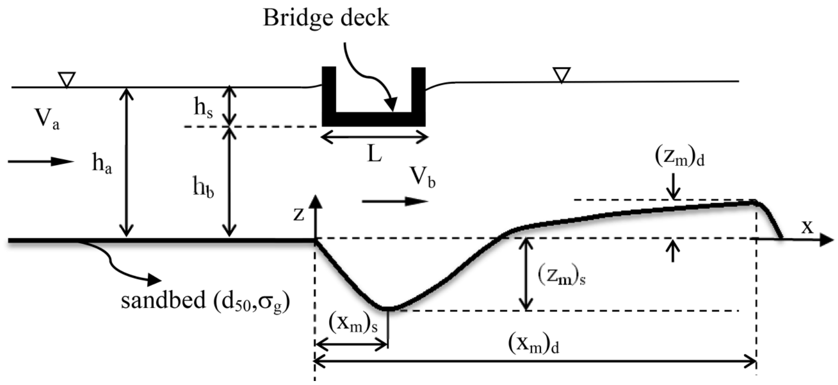
Fig. 1. Definition sketch of scour due to pressurized flow under a bridge deck.



the bridge opening; and V_c is the critical velocity of the bed material (Fig. 1). Arneson (1997) defined critical velocity V_c associated with incipient sediment motion that is based on h_a as

$$(2) \qquad V_c = 1.52\sqrt{g(s-1)d_{50}}\left(\frac{h_a}{d_{50}}\right)^{1/6}$$

in which s is the specific gravity of sediment material; g is the acceleration due to gravity; and d_{50} is the median diameter of sediment particles.

Umbrell et al. (1998) proposed the following equation to estimate pressure scour depth as

$$(3) \qquad \frac{z_s}{h_a} = 1.102\left(\frac{V_a}{V_c}\right)^{0.603} - \frac{h_b}{h_a}$$

where V_a is the mean upstream approach flow velocity. According to Umbrell et al. (1998), it should be noted that V_c is calculated by eq. (2) using the constant of 1.58 rather than 1.52.

Lyn (2008) re-examined HEC-18 and found it to exhibit unsatisfactory behaviour due to ill-chosen original model equations. He proposed a design equation based on a one-sided predictive interval as

$$(4) \qquad \frac{z_s}{h_a} = \min\left[0.21\left(\frac{V_b}{V_c}\right)^{2.95}, 0.6\right]$$

Guo et al. (2009) studied both analytically and experimentally pressure flow under bridges having girders in clear-water conditions. Experiments revealed that (1) equilibrium scour profiles under a bridge are more or less consistent across the channel width; (2) the maximum scour position is located under the bridge and located at a location 15.4% of the deck width from the downstream edge of the deck; and (3) scour begins at approximately one deck width upstream of the bridge, and deposition begins at approximately 2.5 deck widths downstream of the bridge. They also divided analytically bridge scour into three cases: downstream unsubmerged, partially submerged, and totally submerged.

Recently, Zhai (2010) investigated experimentally time-dependent scour depth under bridge decks in totally submerged flow conditions using two uniform sediment sizes and one model bridge deck with three different inundation levels. A semi-empirical model for estimating time-dependent scour was then presented based on the mass conservation of sediment, which agrees well with the collected data.

All previous research given above on pressure-scour covered only steady flow conditions using unique peak discharge value. However, the general practice of employing peak flow discharge to evaluate the maximum scour depth for design may be questioned because the maximum scour depth occurring under a flood hydrograph can be much smaller than that calculated using peak-flow discharge. In other words, using the peak-flow discharge for design can greatly overestimate the maximum scour depth in comparison to the actual conditions under the flood hydrograph. Therefore, when the flow unsteadiness is pronounced, the temporal effect on scour depth should be considered (Chang et al. 2004; Lu et al. 2008).

Besides there is a limited number of studies on bridge scour under pressure flow conditions, the cases considering flow unsteadiness were never considered in the previous studies. Oliveto and Hager (2005) stated that although bridge foundations fail mainly during floods, the effect of unsteady hydraulic load on bridge piers and abutments has received no systematic attention.

Compared to the scour research studies on piers and abutments, pressure-flow scour (sometimes called as vertical contraction scour) has received less attention scientifically. The main goal of this paper is to study the pressurized flow scour under a bridge deck experimentally, governed by both steady and unsteady clear-water flow conditions.

Theoretical considerations

Non-dimensional equilibrium (or ultimate) pressure-scour depth under a bridge deck as given in Fig. 1 can be expressed as a function of the following non-dimensional form of relevant parameters provided that clear-water conditions at the upstream of the bridge deck, single uniform sand size, and constant bridge length along the streamwise direction:

$$(5) \qquad \frac{z_s}{h_a} = f\left(\frac{h_b}{h_a}, \frac{V_b t}{L}, \frac{V_b}{V_c}\right)$$

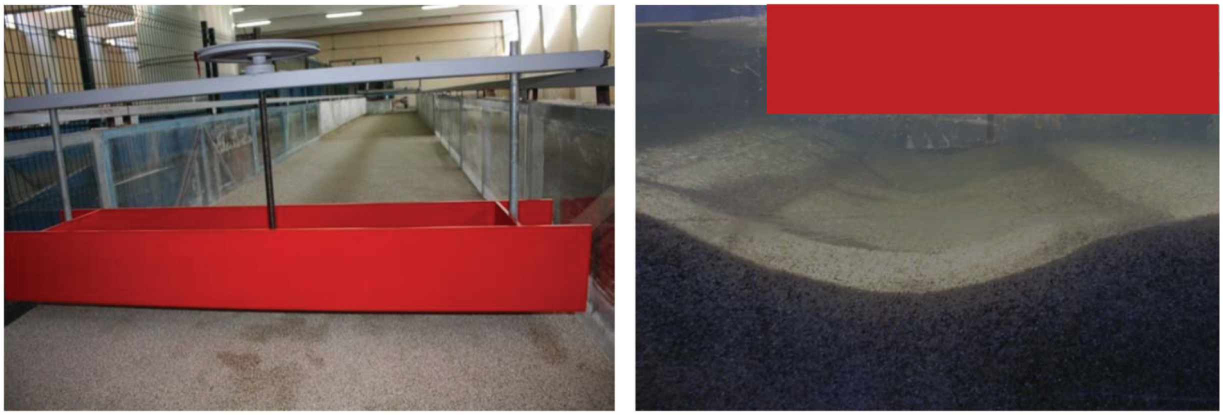
where h_a, V_b, V_c, h_b, and L, are as previously defined and t is time. Herein, h_b/h_a is the submergence ratio of the bridge deck, $V_b t/L$ is the non-dimensional time parameter, and V_b/V_c is the flow intensity under the bridge deck. In this study the non-dimensional parameter representing temporal scour development $V_b t/L$ is replaced with another non-dimensional time parameter of t/τ as given by Hahn and Lyn (2010) in which τ is a time scale of scour development and will be defined in detail in the proceeding sections. Then, eq. (5) can be re-arranged as

$$(6) \qquad \frac{z_s}{h_a} = f\left(\frac{h_b}{h_a}, \frac{t}{\tau}, \frac{V_b}{V_c}\right)$$

Experimental conditions

The experiments were performed at the Hydraulics Laboratory of State Hydraulic Works (DSI) of Turkey, Ankara. A rectangular

Fig. 2. (*a*) Photograph of test channel and bridge deck model used in experimental study (looking from upstream) and (*b*) scour profile under bridge deck section during a test. [Colour online.]



a) b)

flume with transparent walls, 30 m long and 1.5 m wide, was filled with erodible uniform sediment to conduct the experiments (Fig. 2). Sand was used as the sediment material with median size of d_{50} = 0.90 mm and sediment uniformity coefficient of σ = 1.29. The flume slope was 0.001 and a recessed section, 7 m long, containing sediment was located 13 m downstream of the flume inlet. The flume flow was supplied from a constant-head water tank and measured by a sharp-crested rectangular weir. The outflow was directed to the laboratory return channel, from where it was pumped to the water tank for re-supply. The flow depth was read with a conventional point gage, typically to ± 0.5–1.0 mm, depending on the local water surface turbulence. The sand layer thickness was 0.50 m in the working reach and 0.10 m along the rest of the channel. At the end of each test the channel was drained gradually without causing any disturbance to the scour pattern. After each experiment the extent of the scour hole was measured by a point gage along the centerline of the test channel.

As pressure flow under even a simple bridge deck has not been clarified yet, the localized effect of bridge piers and abutments were not included during the experiments in the present study. The modeled bridge deck was of simple rectangular form having the length in the streamwise direction, L = 0.30 m as shown in Fig. 2 without considering a real bridge structure with bottom girders. The deck elevation was adjustable permitting the deck to have two bridge opening heights of h_b = 5 and 7.5 cm. For these two h_b values, four discharges were applied in the experiments (Q = 40, 45, 50, and 55 L/s with the corresponding mean upstream approach velocities varying between V_a = 0.232 and 0.286 m/s). As the maximum scour often results from clear-water flow with a critical approach flow velocity for bed-load motion, experiments were conducted under clear-water flow conditions for both steady and unsteady flows. Next, steady and unsteady flow results were compared with each other. Equation (2) was used to determine whether live-bed or clear-water conditions existed upstream of the bridge deck section. The critical flow velocity V_c was calculated from eq. (2) as V_c = 0.414 m/s, which corresponds to a maximum allowable velocity in the upstream of the test channel. The maximum test duration was 32 h of continuous run for the experiments that were conducted under clear-water steady flow conditions. The data for h_b = 7.5 cm and Q = 40 L/s was excluded from the experimental study since the flow conditions below the bridge deck was below the critical condition for sediment motion.

In unsteady flow conditions, the prescribed stepwise discharge hydrographs as given in Fig. 3 were applied to maintain unsteady clear-water flow conditions in the test channel. The scour pattern, which developed in the test section, was obtained by performing an 8 h continuous run under clear-water flow conditions. In each hydrograph, the peak discharge was set to 50 L/s while time to peak t_p and step-size of hydrographs were varied. Since previous studies showed that the recession limb of flood hydrograph plays a minor role in the scouring process, only the effect of the rising limb of a stepwise hydrograph is addressed in this study (Chang et al. 2004; Oliveto and Hager 2005; Lai et al. 2009; Hager and Unger 2010). The experimental conditions used in the present study are given in Table 1.

Analysis of results

Steady flow

Temporal evolution of bed morphology

Pressurized scour mechanism starts when the vertical constriction of the flow area by the bridge deck with decreasing cross-sectional area and increasing the corresponding flow velocity through the opening. Any increase in flow velocity attributes additional shear stress on the channel bed causing an increase in bed scour in the area of contraction region. As bed material is transported out of the contraction region, the cross-sectional area increases so that the velocity decreases. Eventually, the flow velocity falls below the threshold velocity of the bed material and scour equilibrium is reached.

Horizontal coordinates, x and y, are normalized by the width of the bridge L, so that x/L = 0 and 1 designate the streamwise beginning and end of the model bridge. The vertical coordinates are scaled with the depth of submergence h_s.

The temporal developments of the scour holes in steady flow conditions are shown in Figs. 4 and 5 for h_b = 0.075 m and 0.050 m, respectively, where streamwise profiles of bed elevations for various values of t are plotted. The profiles shown were all measured along the centerline of the test channel, since the maximum measured scour depth $z_s = (z_m)_s$ had minimum side effects.

In general, it can be said that the scour geometry is 2D and the shapes of the scour holes are almost identical as time elapses. It can be also observed that the scour hole develops rapidly from t = 0 to 0.5 h, which means that the rate of change of scour depth is very high at the beginning of scour process. However, the change of scour depth is negligible after t = 30 to 32 h, which implies that an equilibrium scour hole is attained approximately at t = 32 h. The position of the maximum scour depth is near to the downstream part of the bridge deck. Figure 4 indicates that the limit of scour hole develops between $-0.5 < x/L < 1.5$ for $h_b/h_a \approx 0.60$. On the other hand, for $h_b/h_a \approx 0.40$ as given in Fig. 5, limits of scour hole extends further downstream and develops approximately

Fig. 3. Stepwise hydrographs used in the experimental study.

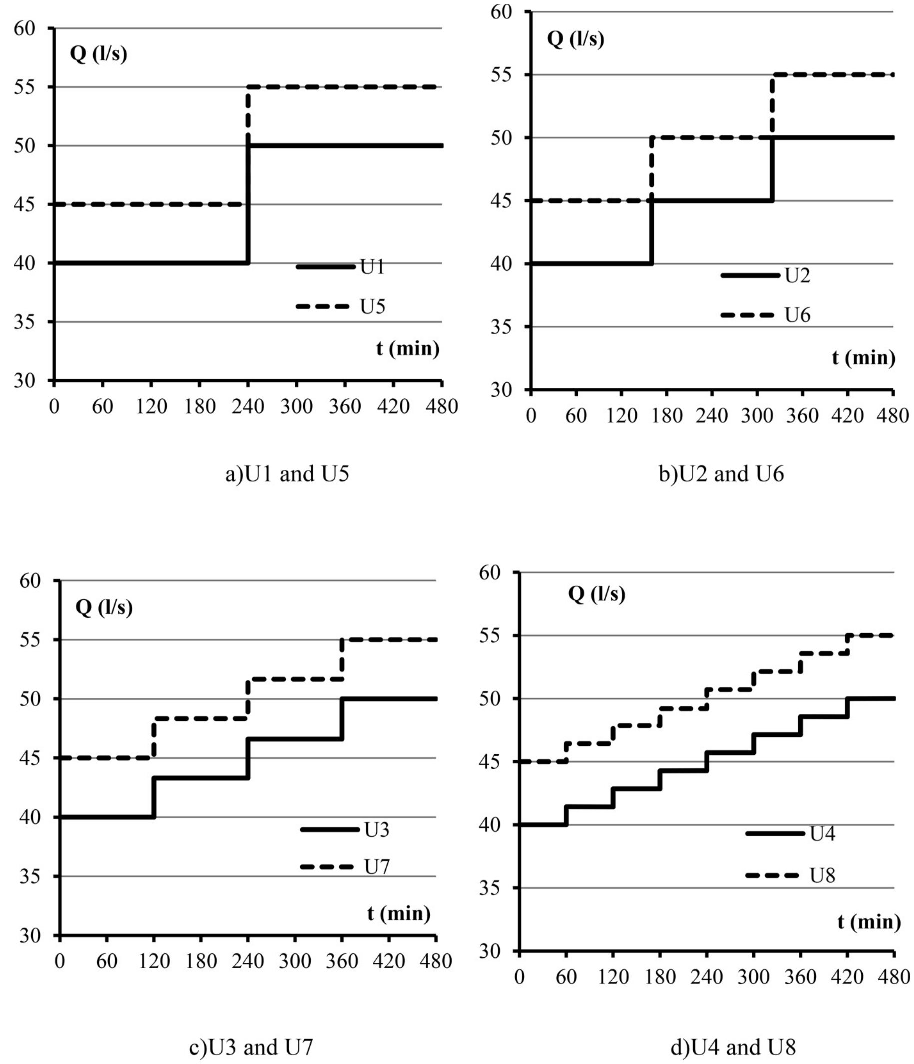


Table 1. Flow, sediment, and geometrical conditions used in experimental study.

Parameter	Steady flow	Unsteady flow
V_a (m/s)	0.232–0.286	0.222–0.266
h_b (m)	0.050, 0.075	0.075
h_a (m)	0.115–0.131	0.120–0.125
h_s (m)	0.045–0.078	0.045–0.050
L (m)	0.30	0.30
d_{50} (mm)	0.90	0.90
σ	1.29	1.29
T (h)	32	8
Fr_a	0.23–0.25	0.20–0.24
h_s/d_{50}	50–87	55

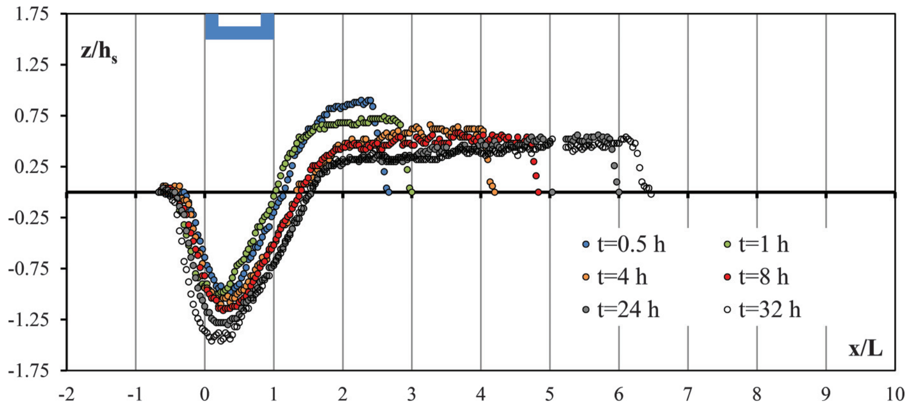
Note: T is maximum test duration (h); Fr_a is approach flow Froude number defined as $Fr_a = V_a/(gh_a)^{0.5}$.

between $0 < x/L < 3.5$. Similar scour profiles were obtained from Guo et al. (2009) for a girder type bridge deck model indicating that the scour occurred generally between $-1.5 < x/L < 1.5$. For both studies, i.e., the present study and Guo et al. (2009), the occurrence of the maximum scour depths was observed under the bridge deck section ($0 \le x/L \le 1$) for all test durations studied. However, in the study of Hahn and Lyn (2010), the scour hole development occurred next to the section downstream of the bridge deck ($x/L > 1$). Comparing Figs. 4 and 5 on the basis of bridge height h_b, it can be easily observed that as the bridge opening height h_b decreases then V_b/V_c increases and as a result scour hole geometry moves along the downstream direction.

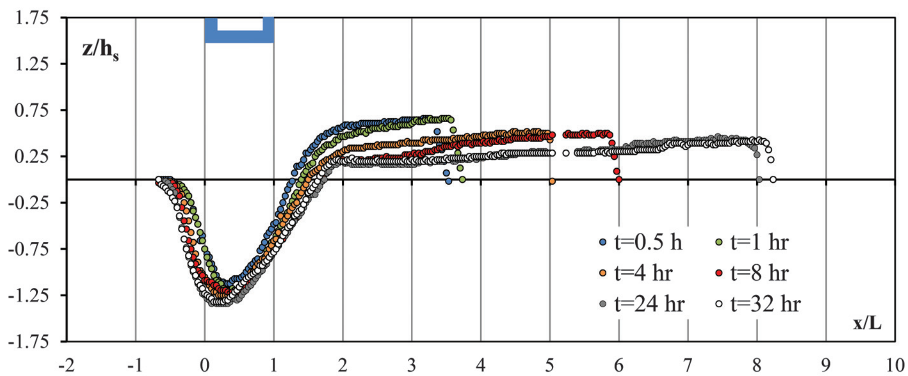
Depositional patterns of the scoured sediment material propagating dune-like bed forms along the downstream direction were also measured in all tests. The sediments eroded from the scour hole were deposited and as a consequence the elevation of the channel bed increased. Similar to the scour profiles, 2D dune-like bed forms downstream of the bridge deck due to pressurized-scour process as a component of scour were observed along the depositional patters. In Fig. 4, the extent of deposition zone developed between $1.5 < x/L < 8.1$ for $h_b/h_a \approx 0.60$ is presented. As shown in Fig. 5, when the velocity under the bridge section increases, the deposition zone extends further downstream (i.e., $3.5 < x/L < 10.7$) for $h_b/h_a \approx 0.40$.

Figure 6 displays the temporal evolution of the maximum scour depth $(z_m)_s$ under the bridge deck as a function of approach flow intensity ratios as $V_a/V_c = 0.64$ and 0.69 for constant bridge heights of h_b = 0.050 and 0.075 m, respectively. For those cases, as time elapses, the maximum scour depth increases but the rate of change decreases and tends to zero as an equilibrium scour approaches. It is also observed from Fig. 6 that as approaching flow intensity increases the maximum scour depth $(z_m)_s$ increases as well. In addition, the scour depth increases as the bridge height, h_b, decreases, or deck inundation level h_a–h_b increases. Figure 7

Fig. 4. Temporal development of scour hole and deposition profiles for h_b = 7.5 cm under steady flow conditions of (*a*) Q = 50 L/s and (*b*) Q = 55 L/s. [Colour online.]



a)



b)

shows temporal evolution of the maximum scour depth $(z_m)_s$ as a function of flow intensity ratios under bridge deck as V_b/V_c = 1.60 and 1.07 for the constant approaching flow intensity V_a/V_c = 0.64 and V_b/V_c = 1.77 and 1.18 for the constant approaching flow intensity V_a/V_c = 0.69, respectively. For a given constant V_a/V_c the maximum scour depth increases with the increase of flow intensity under the bridge deck V_b/V_c. Consequently, Figs. 6 and 7 indicate that the maximum scour depth $(z_m)_s$ increases when either of V_b/V_c or V_a/V_c increases, provided that V_b is the main dominating flow parameter contributing to the scour phenomenon which should be higher than V_c to initiate the scour process.

To analyze the variation of non-dimensional scour parameters $(z_m)_s/h_s$ and $(x_m)_s/h_s$ with non-dimensional time parameter t/τ, time parameter τ should be initially defined. In this study, time scale of the scour development was calculated from the equation of Hahn and Lyn (2010)

$$(7) \qquad \tau = \frac{h_s^2}{\theta_u^{n_s}\sqrt{g(s-1)d_{50}^3}}$$

where θ_u is Shields parameter upstream of the bridge structure $\theta_u = u_*^2/[g(s-1)d_{50}]$, n_s is a constant, and u_* is shear velocity.

The constant n_s varies between 1.5 and 2.5 and can be selected as 2.5 in this study to obtain the likely time scale value, i.e., τ = 4.57 h. The semi-log plot of non-dimensional time scale t/τ versus $(z_m)_s/h_s$ given in Fig. 8 indicates that the maximum scour depth for a given time, $(z_m)_s$, varies logarithmically with time over the range of durations studied as similar to the results obtained in Hahn and Lyn (2010). Regression analysis of the data of the present study for non-dimensional scour depth $(z_m)_s/h_s$ led to the following equation:

$$(8) \qquad \frac{(z_m)_s}{h_s} = 1.28\left(\frac{t}{\tau}\right)^{0.05}$$

with a coefficient of determination of R^2 = 0.69.

The location of the maximum scour depth $(x_m)_s$ is almost constant and for V_b/V_c = 1.07 and 1.18 it slightly varies around the value of $(x_m)_s/L \approx 0.2$ as depicted from Fig. 9. Nevertheless, the location moves about $(x_m)_s/L \approx 1.0$ for V_b/V_c = 1.60 and 1.77 indicating significance of flow intensity and turbulence under the bridge deck on the scour depth location.

The maximum deposition height $(z_m)_d/h_s$ decreases with time. Figure 10 shows the general trend for all data used in the analysis. Except for the low values of non-dimensional time parameter of t/τ, it is evident that data of the different flowrates and bridge heights conform to a narrow band. Higher scatter on dune-like deposition heights can be observed at low t/τ values, which are mainly due to flow instabilities resulted from the scour hole turbulence effect on dune-like bed forms at the start of scour-deposition process. This effect is limited as time elapses. The general trend shown by the figure implies that they can be represented by a single curve given by

$$(9) \qquad \frac{(z_m)_d}{h_s} = 0.52\left(\frac{t}{\tau}\right)^{-0.078}$$

Fig. 5. Temporal development of scour hole and deposition profiles for h_b = 5 cm under steady flow conditions of (*a*) Q = 50 L/s and (*b*) Q = 55 L/s. [Colour online.]

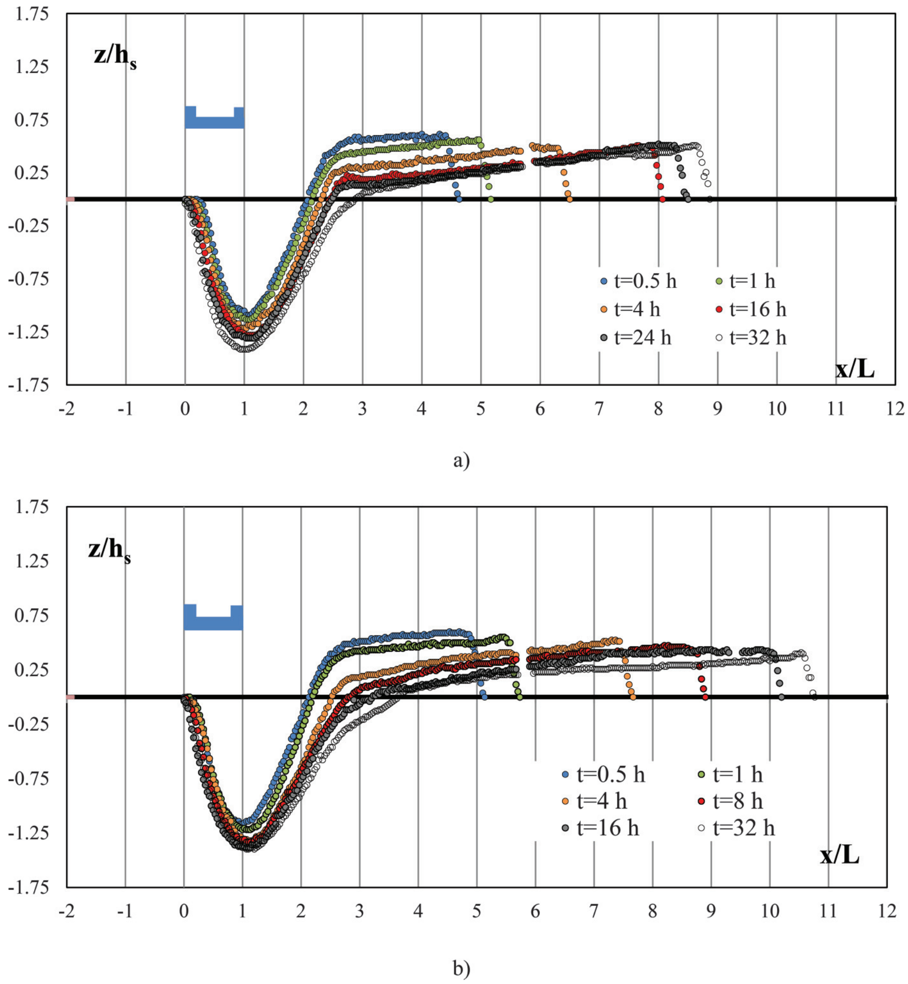
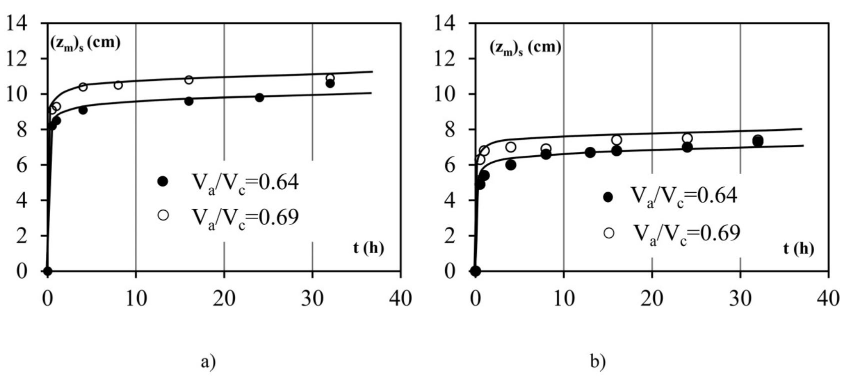


Fig. 6. Developments of the maximum scour depths under clear-water steady flow conditions for constant bridge heights of (*a*) h_b = 5 cm and (*b*) h_b = 7.5 cm.



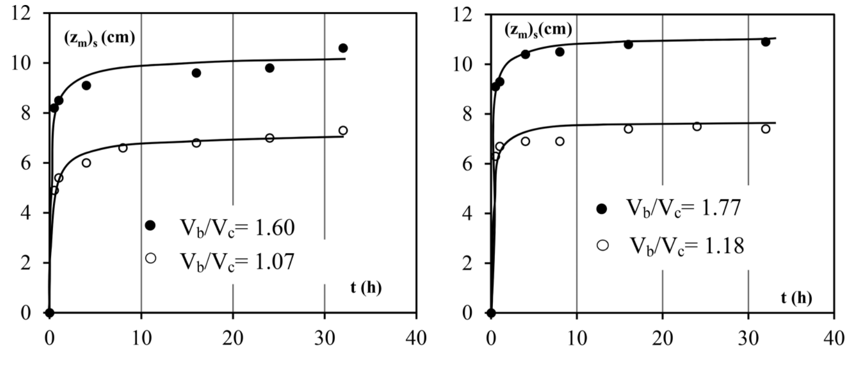
with a relatively low coefficient of determination of R^2 = 0.42.

Figure 11 shows that the location of the non-dimensional maximum deposition height $(x_m)_d/L$ varies with non-dimensional time t/τ over the range of experimental conditions. Flow intensity under the bridge deck V_b/V_c is significant on the non-dimensional deposition location. The location of the deposition zone moves further downstream with higher flow intensities as shown in Fig. 11. It can be inferred from Fig. 11 that $(x_m)_d/L$ values vary between 7 and 10.5 at the end of the test duration within the limits of experimental conditions applied.

Design equation for $(z_m)_s$

Data used in the analysis of the maximum scour depth $(z_m)_s$ cover data obtained from this study together with the data published by Arneson (1997) and Umbrell et al. (1998). Figure 12 shows the plot of non-dimensional parameters of $[h_b + (z_m)_s]/h_a$ versus V_b/V_c for all data used in the analysis. For consistency, the pier and overflow cases were excluded from Arneson's and Umbrell's data set, respectively. Applying any regression analysis model to include all data sets leads to under-prediction of the observed data due to the high scatter nature of data used mainly as a result of

Fig. 7. Developments of the maximum scour depths under clear-water steady flow conditions for constant approaching flow intensities of (*a*) V_a/V_c = 0.64 and (*b*) V_a/V_c = 0.69.



a) b)

Fig. 8. Variation of non-dimensional maximum scour depth parameter $(z_m)_s/h_s$ with non-dimensional time scale t/τ under clear-water steady flow conditions.

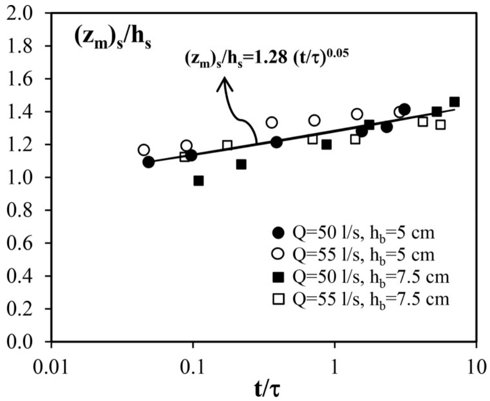


Fig. 9. Variation of non-dimensional location of maximum scour depth parameter $(x_m)_s/L$ with non-dimensional time scale t/τ under clear-water steady flow conditions.

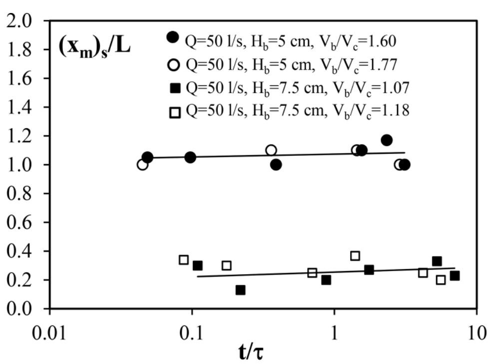


Fig. 10. Variation of non-dimensional maximum deposition height parameter $(z_m)_d/h_s$ with non-dimensional time scale t/τ under clear-water steady flow conditions. [Colour online.]

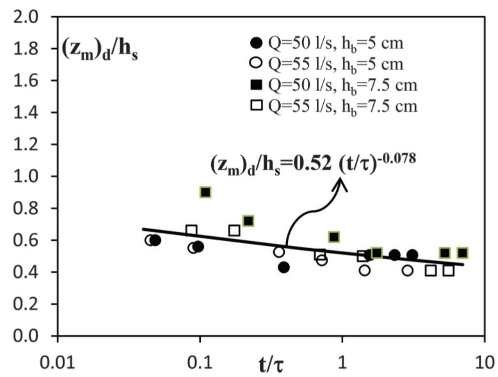
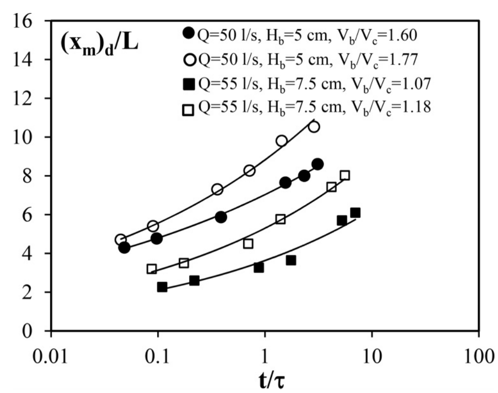


Fig. 11. Variation of non-dimensional location of maximum deposition height parameter $(x_m)_d/L$ with non-dimensional time scale t/τ under clear-water steady flow conditions.



unclear or insufficient experimental durations to obtain the maximum scour depth. Lyn (2008) also criticized the test durations of Arneson (1997) and Umbrell et al. (1998) data sets in this respect. In Fig. 12, it should be noted that vertical contraction effect on scour mechanism is considered with both flow intensity under the bridge deck section V_b/V_c and bridge submergence (without overflow) h_b/h_a. When upper envelope lines are drawn on Fig. 12 passing through the maximum $[h_b + (z_m)_s]/h_a$ values, two linear tendencies can be observed on the basis of V_b/V_c; i.e., $V_b/V_c < 1$ and $V_b/V_c \geq 1$. The envelope plotted leads to a set of simple linear equations as follows (eq. 10) that can be used for predicting design depth of scour in a pressurized flow under bridge decks. Equation (10) includes the effect of h_b/h_a on scour mechanisms and is simple in application to find a value for design purposes.

$$\frac{[h_b + (z_m)_s]}{h_a} = 0.65 + 0.5\left(\frac{V_b}{V_c}\right) \quad \text{for } 0.5 \leq \frac{V_b}{V_c} < 1 \tag{10a}$$

$$\frac{[h_b + (z_m)_s]}{h_a} = 1.025 + 0.125\left(\frac{V_b}{V_c}\right) \quad \text{for } 1 \leq \frac{V_b}{V_c} \leq 1.8 \tag{10b}$$

Equation (10) is valid for the conditions under which the experiments were conducted in this study and in earlier studies.

Fig. 12. Variation of bridge submergence with flow intensity.

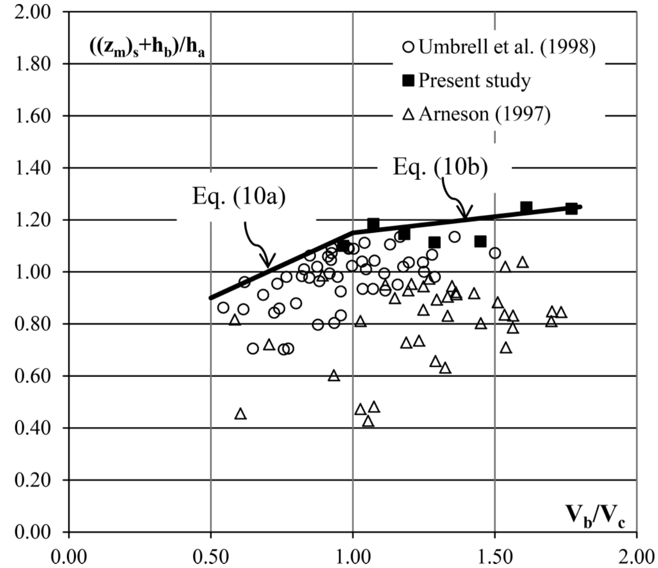


Fig. 13. Effect of t_p/T on (*a*) the maximum scour depth and (*b*) the location of the maximum scour depth for h_b = 7.5 cm.

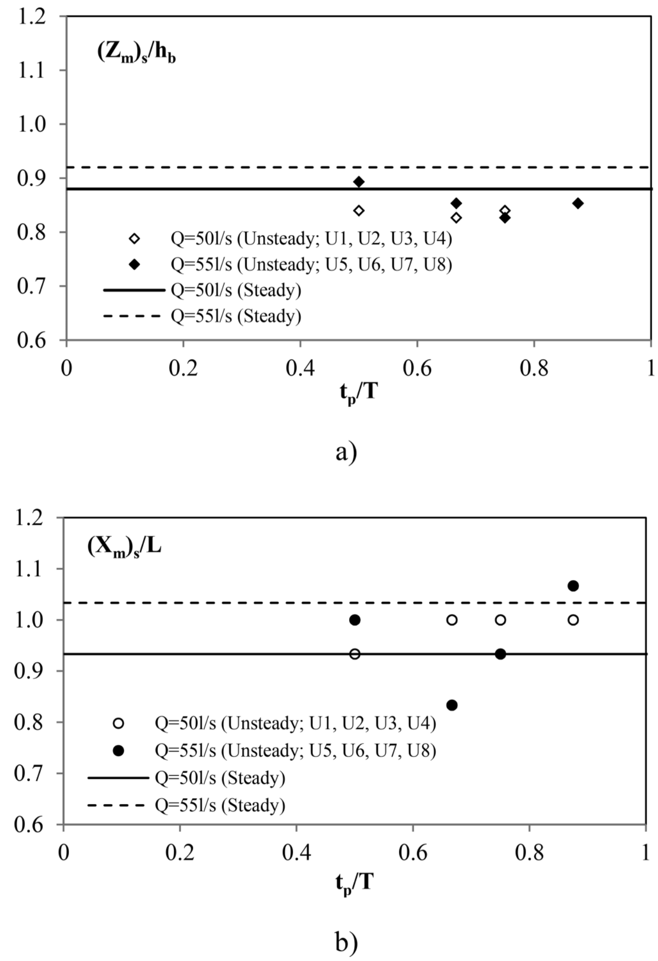
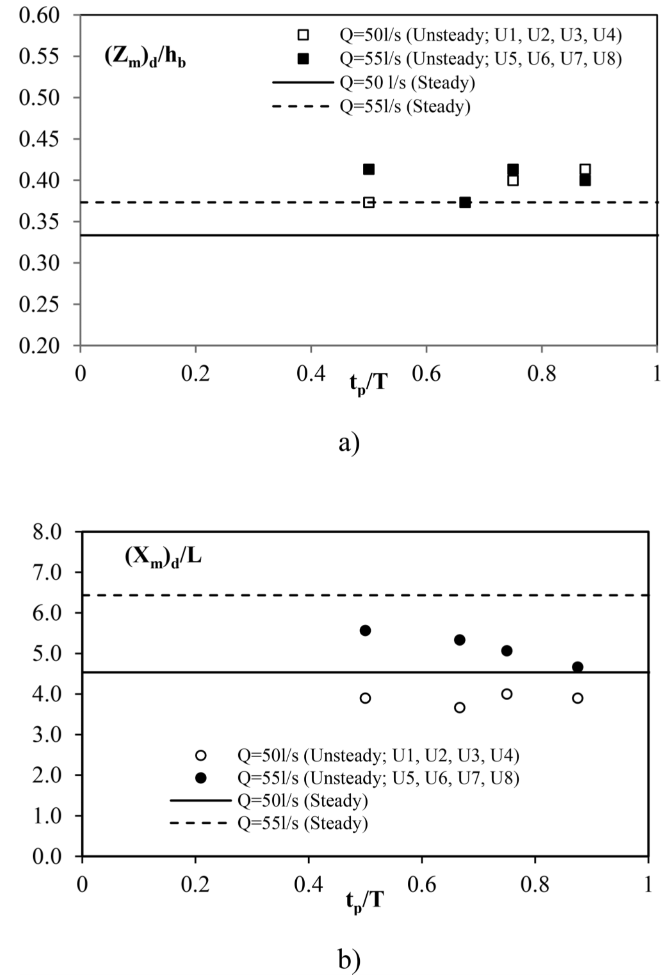


Fig. 14. Effect of t_p/T on (*a*) the maximum deposition height and (*b*) location of the maximum deposition height for h_b = 7.5 cm.



Unsteady flow

In this section, the unsteady flow (stepwise hydrographs) experimental results are discussed in detail. The effect of time to peak (t_p) on scour and dune-like deposition under various hydrographs are investigated. In the unsteady flow experiments, general observations show that the scour and deposition profiles are self-similar. Unsteady flow experiments characterized by the stepwise hydrographs were compared with the steady flow case of maximum peak discharge Q_p = 50 L/s and 55 L/s with T = 8 h. The scour and dune-like depositional profiles developed at the end of peak-flow discharges are measured for all stepwise hydrographs (U1–U8 as given in Fig. 3) with peak-flow velocities of $(V_a)_p$ = 0.266 m/s and $(V_a)_p$ = 0.286 m/s corresponding to Q_p = 50 L/s and Q_p = 55 L/s, respectively, but different time to peak values of t_p = 240, 320, 360, and 420 min.

The final (end) scour depths for almost all unsteady tests are less than the maximum scour depths for the steady flow conditions (Fig. 13*a*). Similar to steady flow conditions, higher flow intensities or peak discharge values (Q_p) cause higher maximum scour depths in unsteady flow conditions. In addition, effect of time to peak discharge, t_p, on the maximum scour depth was limited for all cases. As expected, effect of t_p on scour depth evolution is not of great importance as was similarly found in the studies on pier scour under flood waves (Chang et al. 2004; Schillinger and Khan 2013). Experimental results indicate that multiple smaller flood events do not cause greater scour depth than the larger flood events that cause equilibrium scour depths. Figure 13*b* shows that for all stepwise hydrographs studied in the unsteady flow conditions the location of the maximum scour depths are slightly less than the scour depths obtained from the steady flow cases.

Profiles of dune-like bed forms along the downstream of the bridge deck are also measured in the unsteady flow conditions. Variations of t_p/T with maximum deposition height normalized by bridge opening height (i.e., $(z_m)_d/h_b$) and its corresponding location normalized by bridge width (i.e., $(x_m)_d/L$) are plotted in Figs. 14*a* and 14*b*, respectively. It can be inferred from Fig. 14*a* that the maximum deposition heights in unsteady cases are higher than the steady flow deposition heights. Approaching flow intensity V_a/V_c and time to peak ratio t_p/T have limited effect on the maximum deposition heights. In Fig. 14*b*, it is observed that locations of the maximum deposition heights in unsteady flows are less than the values of steady flow conditions. Comparing with the steady flow conditions, the locations are 10%–30% less in the unsteady cases showing a trend of decrease as the flow intensity decreases.

Conclusions

An experimental study was conducted using various experimental combinations with a duration of up to 32 h to investigate the temporal variation of scour progress and downstream sediment deposition in steady flow conditions. A single sediment size, four different approach flow velocities corresponding to degree of bridge deck submergences, and two bridge opening heights were used in the experimental program. Moreover, for the unsteady flow conditions, four stepwise hydrographs were studied to analyze flow unsteadiness on pressure-scour depth and downstream deposition profiles. Within the range of experiments, the following conclusions can be drawn from the results of the present experimental study:

(1) Scour and dune-like deposition profiles are 2D and self-similar as time elapses in steady flow conditions and four stepwise hydrographs (unsteady flow) studied.
(2) Under steady flow experiments, limits of the scour hole developed covered a horizontal distance of $-0.5 < x/L < 1.5$ for $h_b/h_a \approx 0.6$ and $0 < x/L < 3.5$ for $h_b/h_a \approx 0.4$ at the end of the test duration. The location of the maximum scour depth $(x_m)_s$ is almost constant and slightly varies around the value of $(x_m)_s/L \approx 0.2$ for $h_b/h_a \approx 0.6$. Nevertheless, the location moves about $(x_m)_s/L \approx 1.0$ for $h_b/h_a \approx 0.4$.
(3) The extent of the deposition zone developed between $1.5 < x/L < 8.1$ for $h_b/h_a \approx 0.6$. The deposition zone extends further downstream for $h_b/h_a \approx 0.4$.
(4) Equation (10) is proposed to estimate a design value for predicting the maximum scour depth z_s based on the analysis of data from the experiments as well as from the earlier studies. The derivation of eq. (10) involves two important parameters: flow intensity under the bridge deck section V_b/V_c and bridge submergence h_b/h_a.
(5) The scour depths for almost all unsteady tests are less than the maximum scour depths obtained from the steady flow conditions. Effect of time to peak t_p on scour development is limited in unsteady flow conditions. The maximum deposition height locations are 10%–30% less in unsteady cases when compared with the steady cases.

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List of symbols

d_{50}	median diameter of sediment particles
g	gravitational acceleration
h_a	upstream flow depth
h_b	distance between the bridge low chord and the undisturbed bed at the bridge deck section
L	width of bridge
Q	discharge
R	coefficient of determination
S	slope of the test channel
s	specific gravity of sediment material
t	time
t_p	time to peak discharge
T	test duration
V_a	mean upstream approach velocity
V_b	initial (prior to scour) velocity through the bridge opening
V_c	critical velocity of the bed material associated with incipient sediment motion
z_s	depth of equilibrium or ultimate scour
ρ	density of the water
ρ_s	density of the sediment
σ_g	standard deviation of sediment size distribution
τ	non-dimensional time parameter